O-F-371B
February 2, 1979
SUPERSEDING
O-F-371a
October 31, 1966

FEDERAL SPECIFICATION

FIRE EXTINGUISHING AGENT, SODIUM BICARBONATE, DRY CHEMICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal Agencies.

1. SCOPE

1.1 This specification covers the requirements for a dry chemical fire extinguishing agent powder composed largely of sodium bicarbonate.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-D-705 - Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters).
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards, at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government Activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C204 - Fineness of Portland Cement by Air Permeability Apparatus.
D5 - Penetration of Bituminous Materials.
E11 - Wire-Cloth Sieves for Testing Purposes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Fire Protection Association (NFPA) Standard:

No. 10 - Portable Fire Extinguishers.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Underwriters Laboratories, Inc., Standards:

UL299 - Dry Chemical Fire Extinguishers.
UL711 - Rating and Testing of Fire Extinguishers.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish the material, in quantities specified, for first article inspection and approval (see 4.2.1 and 6.3).

3.2 Material. The dry chemical agent shall be as specified herein. The material shall be approved for use in fire extinguishers conforming to UL299, and specified by make and model (see 6.2), in accordance with the requirements of NFPA No. 10 for recharge chemicals.

3.2.1 Conformance to UL requirements. The contractor shall submit to the contracting agency proof that the fire extinguishing agent he proposes to supply under the specification conforms to the requirements of UL299 for dry chemical extinguishing agents used with the specified extinguishers. The label or listing with re-examination of UL299 may be accepted as evidence that the fire extinguishing agent conforms to this requirement. In lieu of such label or listing, the contractor may submit independent proof, satisfactory to the contracting agency, that the fire extinguishing agent conforms to applicable requirements, including methods of tests, and including charged extinguisher performance tests, of UL299. Compliance with this requirement does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his product.

3.2.2 Chemical and physical requirements. The material shall conform to the chemical and physical requirements of table I. Materials may be added to improve the fluidity, water repellency, and noncaking characteristics of the finished agent. These additives shall not give off toxic vapors when subjected to intense heat.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:13Z
Check the source to verify that this is the currentversion before use.

Table I. Chemical and physical requirements.

Requirement	Value	Test para.
Sodium bicarbonate ($NaHCO_3$), minimum, mass percent	90.0	4.5.1
Sodium carbonate (Na_2CO_3), maximum, mass percent	2.0	4.5.1
Inert, water-insoluble ingredients, maximum, mass percent	8.0	4.5.1
Moisture, maximum, mass percent	0.20	4.5.2
Apparent density, g/cm^3	0.90-1.10	4.5.3
Packed density, minimum, g/cm^3	1.35	4.5.4
Caking and lumping, minimum penetration, mm	15.0	4.5.5
Packed powder discharge, minimum, mass percent	90.0	4.5.6
Specific surface area, cm^2/g	2000-4000	4.5.7
Color identification	Natural 1/	4.5.8

1/ Without added coloring agents.

3.2.2.1 Particle size. The particle size distribution of the material shall conform to the requirements of table II, when determined as specified in 4.5.9.

Table II. Particle size requirements.

U.S.A. Standard Sieve	Percent retained 1/	
	Minimum	Maximum
40	-	0.0
100	0.0	5.0
200	0.0	10.0
325	5.0	20.0
Bottom pan	75.0	95.0

1/ The total percent retained on the 100, 200, and 325 screens shall not exceed 25.0.

3.2.2.2 Low temperature characteristics. At -65° Fahrenheit (F) (-53.9° Celsius (C)), the material shall be free-flowing with no caking, when tested as specified in 4.5.10.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on the material in quantities specified in 4.4, when a first article sample is required (see 3.2 and 6.2). This inspection shall include the testing of 4.5. Failure of the first article to pass any of the tests shall be cause for rejection. The first article may be standard production material from the contractor's current inventory provided the material meets the requirements of the specification and is representative of the manufacturing technique applicable to the remaining material to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample material selected in accordance with 4.3 and 4.4. This inspection shall include the testing of 4.5.

4.3 Inspection lot. An inspection lot shall consist of five production batches or less when this fulfills the contract. When more than five batches are required, the batches shall be grouped as equally as possible by weight to give not less than three nor more than five lots. A production batch shall consist of material from a single mix or run, manufactured under essentially the same conditions and at essentially the same time.

4.4 Sampling. Lot samples of raw ingredients will be required when batch sheets are furnished, as specified in 4.5.1.2. The lot sample for the tests of 4.5.1 through 4.5.5, and 4.5.7 through 4.5.10, shall be a composite sample, representing the content and proportionate amount of each batch, and shall consist of not less than 15 pounds (lb) (7 kilograms (kg)) total sample. An additional three each composite samples shall be similarly drawn from the lot batches for the test of 4.5.6 as specified therein. The samples shall immediately be placed in clean, dry, airtight and watertight containers, sealed, and marked with material identification, lot number, and date.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:13Z
Check the source to verify that this is the current version before use.

4.5 Testing. Samples selected in accordance with 4.4 shall be tested for conformance to the requirements of section 3. Each test shall be conducted three times, and the resulting values averaged and reported. If the test samples fail to meet all the requirements of section 3, the lot represented shall be rejected.

4.5.1 Composition. Determine the composition of the material by analytical methods (see 4.5.1.1) or by providing batch sheets (see 4.5.1.2).

4.5.1.1 Analytical determination. Determine the sodium bicarbonate and sodium carbonate contents utilizing carbon dioxide loss in a Geissler alkalimeter, or other procedures, together with other appropriate laboratory methods. Determine the inert, water-insoluble ingredients by laboratory methods appropriate for the water-repellant materials present. The full, detailed laboratory procedures, and all individual test values, for the three determinations shall be furnished, as part of the test report, to the Government for approval. The identity of the chemical compounds used as water-repellant additives shall be furnished to the Government by the contractor as part of the test report. When specified (see 6.2), the requirements of 4.5.1.2 shall also be invoked to assist the Government in establishing standard analytical procedures.

4.5.1.2 Batch sheets. Furnish batch sheets showing the complete chemical composition by mass percent as compounded, with manufacturer's lot numbers and certifications for individual raw ingredients, to the Government. Lot samples of raw ingredients, consisting of not less than 2 lb (1 kg) each, shall be furnished for inspection by the Government.

4.5.2 Moisture content. In an atmosphere of 50 percent or less relative humidity and a temperature of 70 ±5°F (21 ±3°C), measure a sample portion of 50 grams ±1 milligram in a tared aluminum dish approximately 68 millimeters (mm) in diameter by 18 mm in depth. Place the dish in a desiccator, using A.C.S. reagent grade sulfuric acid, 95 to 98 mass percent (1.8337 to 1.8361 specific gravity), as a drying agent. Maintain the closed desiccator and contents at a temperature of 70 ±5°F for 24 hours. At the end of this period, redetermine the mass of the sample portion. Express the change in mass in percent as moisture content.

4.5.3 Apparent density. Place a sample portion of 100 grams ±0.1 gram in a clean, dry 250 cubic centimeter (cm^3) glass-stoppered graduated cylinder having an inside diameter (id) of approximately 3.5 cm. Stopper the cylinder and rotate slowly end-over-end for 10 complete cycles. After agitation, set the cylinder upright and allow to settle for 60 seconds. Read the volume occupied by the sample portion to the nearest 0.5 cm^3, and express the density as apparent density.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

4.5.4 Packed density. Procure a conveniently shortened 250 cm^3 graduated glass cylinder, having an id of approximately 3.5 cm, fitted with a No. 8 rubber stopper, and a Tyler Ro-Tap or equal sieve-shaking machine. Measure a sample portion of 100 grams ±0.1 gram in the tared graduate. Place the graduate in a holder modified to fit into the shaking machine, and shake the graduate on the machine for 30 minutes. Read the volume of the packed material to the nearest 0.5 cm^3, from the graduated cylinder, and express the density as packed density.

4.5.5 Caking and lumping test. Place a 125 gram portion of sample in a 100 cm^3 nickel crucible-shaped cup measuring 60 mm in diameter at the top and 64 mm in depth. Vibrate for 5 minutes on a Tyler Ro-Tap or equal sieve shaker using a suitable holder. Place the sample portion in a moving air stream humidifier at 78 percent relative humidity for 24 hours and then place the sample portion in a drying oven at 118°F (48°C) for 24 hours. At the conclusion of the second period, determine the degree of caking with a Precision Universal Penetrometer, or equal, meeting the requirements of ASTM Method D5. Penetration is defined as the resistance of the material, expressed as the distance that a standard needle penetrates vertically into the sample under given conditions of load and time. Use the needle and spindle assembly having a total mass of 50 grams and a time of 5 seconds. Average three readings from the sample portion and express the penetration as mm.

NOTE: A 250-mm diameter desiccator with a saturated ammonium chloride solution in the lower compartment may be used in this determination. During operation, a circulating air supply of 83 cm^3 per second shall be presaturated by bubbling through a saturated ammonium chloride solution and then introduced into the desiccator through a center tube, with 6 mm id, terminating 20 mm above the center hole in the desiccator plate. It is advisable to monitor the relative humidity of the exit air stream.

4.5.6 Packed powder discharge. An approved extinguisher (see 3.2), of make and model as specified (see 6.2), shall be used for determination of percent discharge. Unless otherwise specified (see 6.2), a 10 lb cartridge-operated design shall be used.

4.5.6.1 Procedure. Introduce 10 lb ±2 ounces of material, or an appropriate amount for a designated extinguisher of other capacity into the tared fire extinguisher. Vibrate the charged extinguisher at a frequency of 34 cycles per second and at an excursion of 0.050 inch for a period of 30 minutes. Place the extinguisher in an oven for 24 hours at 130 ±5°F (54 ±3°C). Upon completion of the oven cycle, pressurize the extinguisher with the proper CO_2 cartridge. After maximum pressure has been achieved, discharge the extinguisher until the pressure is dissipated. Redetermine the mass of the extinguisher, and express the packed powder discharge in percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

4.5.7 Specific surface area. Specific surface area shall be determined in accordance with ASTM Method C204 except as specified herein. Modify the apparatus 1/ by doubling the test bed depth, in order to increase the time interval, by machining down the plunger to give an effective bed depth of 2.93 cm instead of the original 1.47 cm. The adjustment of the depth yields an air flow time of from 30 to 360 seconds with various powders. Determine the density of the material by using a 50 cm^3 pycnometer with n-heptane as the liquid. Use a sample portion of a mass producing a bed having a porosity of 0.46. To insure duplicate filling of the cell, make successive additions of 1 gram each, lightly tapping or vibrating after each addition. The cell shall be so filled that the plunger can be depressed to the collar when a pressure of 10 lb (4.54 kg) is applied for a period of 180 seconds. Use Carman's equation, relating specific surface to permeability and utilizing the constants of the apparatus, for all calculations, including calibration calculations, as follows:

$$S = \frac{14}{\rho\ (1-\varepsilon)} \sqrt{\frac{\varepsilon^3\ A\ \overline{P}\ T}{V\ \eta\ L}}$$

Where S = specific surface area, $cm^2 g$
ρ = density of the powder, g/cm^3
T = time required for volume V of fluid to pass through the powder bed, s
η = viscosity of air at temperature at which the test was conducted (see table III), Pa·s
A = area of bed of powder, cm^2
$\overline{P}$ = mean effective pressure difference across the bed of powder, Pa
V = total measured volume of fluid passing through the powder, cm^3
L = depth of bed of powder, cm
ε = porosity = $1 - \frac{\text{mass of bed material, g}}{\text{volume of bed material, } cm^3 \cdot \rho}$

With the apparatus modified as described above to double the bed depth, the following terms are constant, and take the following values:

$\overline{P}$ = 594.6 Pa
A = 1.267 cm^2
L = 2.93 cm
Volume of bed = 3.71 cm^3
V = 2.13 cm^3

1/ The Blaine air permeability apparatus is available from the Humboldt Mfg. Co., 7300 West Agatite, Chicago, IL 60656.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

Substituting the constant values, the equation reduces to:

$$S = \frac{153.83}{\rho\ (1-\varepsilon)} \sqrt{\frac{\varepsilon^3\ T}{\eta}}$$

Where: $\varepsilon = 1 - \frac{\text{mass of bed material, g}}{3.71\ \text{cm}^3 \cdot \rho}$

TABLE III. Viscosity of air at room temperatures.

Room temperature, °C	Viscosity of air, μPa•s
20	18.08
22	18.18
24	18.28
26	18.37
28	18.47
30	18.57

4.5.8 Color identification. The material sample, viewed in bulk, shall have no color other than the natural color of the ingredients used in the composition of the material, to determine compliance with table I.

4.5.9 Particle size. The shaker used shall operate with a single eccentric circular motion at 285 ±10 revolutions per minute, and with a tapping action of 150 ±5 strokes per minute. Screens used for particle size analysis shall conform to ASTM E11, and shall be stainless steel. In an atmosphere of 50 percent relative humidity or less, place an accurately weighed sample of approximately 50 grams in the top screen of a nested series of 8-inch USA standard screens consisting of a 40-mesh, 100-mesh, 200-mesh, and a 325-mesh, with cover and bottom pan. Agitate the screen nest for 5 minutes on a shaker employing eccentric circular motion and tapping action such as the Tyler Ro-Tap machine. Separate the screen nest and accurately weigh the amount remaining on each screen, expressing the results in percent of the original sample.

NOTE: Care must be taken to remove all the residue from the screen. Sieves should be cleaned with a camel's-hair brush to insure complete removal of the fine particles.

4.5.10 Low-temperature characterisitics. Place approximately 20 grams of the dry chemical agent in a 6-inch stoppered test tube. Lower the temperature to -65 ±2°F (-53.9 ±1.1°C) and allow to remain for 1 hour. Remove the test tube and invert without removing the stopper. The material shall flow freely.

4.6 Inspection of preparation for delivery. The packing and marking shall be inspected to verify conformance to the requirements of section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Pails. Unless otherwise specified (see 6.2), the material shall be packed in 50 lb quantities in pails conforming to PPP-P-704, type II, class 4, except as specified herein. The container shall be provided with a 4-mil polyethylene liner conforming to L-P-378. The open end of the liner shall be sealed in a manner that will prevent entry of moisture. A cover with a lever lock, reusable type ring shall be used in lieu of the lug cover. The ring and cover shall be fitted with a rubber sealing gasket.

5.1.1.2 Drums. When specified (see 6.2), the material shall be packed in larger than 50 lb quantities in drums conforming to PPP-D-705 or PPP-D-729, as applicable, with full removable head with a lever-lock reusable type ring closure, fitted with a rubber sealing gasket. Drum liners shall be as specified in 5.1.1.1.

5.1.2 Commercial. The material shall be packed in quantities specified (see 6.2), in accordance with commercial practice to insure carrier acceptance and delivery at destination, at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Unitization. When shipments to Government depots are full car or truckload, the containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 lb in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.3 Marking. Marking shall be as specified in the contract or order, and as specified in 5.3.1.

5.3.1 Special marking.

5.3.1.1 Extinguisher makes and models. Containers shall be marked with the fire extinguisher make and model numbers for which the dry chemical agent has been approved for recharge as specified (see 6.2).

5.3.1.2 Storage and use. Containers shall be marked with instructions for storage and use: "Store in a dry place, at temperatures below 140°F (60°C). After use, reseal bag and close container tightly."

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:13Z
Check the source to verify that this is the currentversion before use.

6. NOTES

6.1 Intended use. The material covered by this specification is approved for use in recharging dry chemical fire extinguishers, of make and model specified (see 3.2), designed for use with sodium bicarbonate base fire extinguishing agents. It should not be used in dry chemical fire extinguishers designed for use with potassium bicarbonate (Purple K) or multipurpose phosphate base (ABC) materials. The material covered herein is to be used on class B and C fires only, as defined in UL711, and as described for the selection of extinguishers in NFPA No. 10, Portable Fire Extinguishers, and Appendixes A and E, and for general information, as described for the agent characteristics, use, and limitations of dry chemical extinguishing systems in NFPA No. 17, Dry Chemical Extinguishing Systems, and Appendix A.

6.2 Ordering data. Purchasers shall select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) When a first article is required for inspection and approval (see 3.1, 4.2.1, and 6.3).
(c) Extinguisher makes and models in which material is used (see 3.2 and 5.3.1.1).
(d) When batch sheets and ingredient certifications and samples are required with composition test procedures and results (see 4.5.1.1).
(e) Make and model of 10-lb cartridge-operated extinguisher for packed powder percent discharge test, or other size or design if required (see 4.5.6).
(f) Level of packing required (see 5.1).
(g) If packing in pails is other than specified (see 5.1.1.1).
(h) When materials are to be packed in drums (see 5.1.1.2).
(i) Quantity to be packed for commercial (see 5.1.2)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of specified quantities of material. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test, and approval of the first article.

6.4 Requirement for batch sheets. The option of 4.5.1.1, to require batch sheets, should only be invoked if the Government is to establish standard analytical procedures for determining material composition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 99

Review activity:

DLA -CS

User activities:

Navy - CG, MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
TVA

Preparing activity:

Navy - YD

Project No. 4210-0299

Orders for this publication are to be placed with General Services Administration, acting as an agent for the [illegible] of documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.